🎭⏳ Playwright Waiting Methods

> ***Purpose:*** *Playwright automatically waits for most actions. However, in some situations, you may need to explicitly wait for a page, element, event, or condition before continuing with your test.*

📁 Wait Methods Overview

Method	Waits For	Common Usage
`page.waitForLoadState('domcontentloaded')`	HTML document loaded	Wait until DOM is ready
`page.waitForLoadState('load')`	Entire page loaded	Wait for images, CSS, JS
`page.waitForLoadState('networkidle')`	No network requests for 500 ms	SPA applications (use cautiously)
`page.waitForSelector()`	Element appears in DOM	Wait for dynamic elements
`page.waitForTimeout()`	Fixed amount of time	Debugging only
`page.waitForURL()`	URL changes	Navigation verification
`page.waitForFunction()`	JavaScript condition becomes true	Custom waiting conditions
`page.waitForEvent()`	Browser/page event occurs	Downloads, popups, dialogs

1️⃣ page.waitForLoadState('domcontentloaded')

📖 Description

Waits until the browser fires the **DOMContentLoaded** event.

This means:

- ✅ HTML has been completely loaded
- ✅ DOM is ready
- ❌ Images may still be loading
- ❌ CSS/Fonts may still be loading

🎯 When to Use

- Wait until page structure is ready
- Before locating elements
- Faster than waiting for complete page load

✅ Syntax

```
await page.waitForLoadState('domcontentloaded');
```

💻 Example

```
await page.goto('https://example.com');

  await page.waitForLoadState('domcontentloaded');

  await page.locator('#username').fill('admin');
```

2️⃣ page.waitForLoadState('load')

📖 Description

Waits until the browser fires the **load** event.

This means:

- ✅ HTML loaded
- ✅ CSS loaded
- ✅ Images loaded
- ✅ JavaScript files loaded

Entire page is fully loaded.

🎯 When to Use

- When images must be visible
- Screenshot testing
- Visual testing

✅ Syntax

```
await page.waitForLoadState('load');
```

💻 Example

```
await page.goto('https://example.com');

  await page.waitForLoadState('load');

  await page.screenshot({ path: 'homepage.png' });
```

3️⃣ page.waitForLoadState('networkidle')

📖 Description

Waits until there are **no network requests for at least 500 ms**.

Useful for applications that load data using APIs.

⚠️ Note

Playwright recommends **avoiding** `networkidle` in most tests.

Prefer waiting for:

- specific elements
- assertions

- API responses

🎯 When to Use

- Single Page Applications (React, Angular, Vue)
- After API data loading

✅ Syntax

```
await page.waitForLoadState('networkidle');
```

💻 Example

```
await page.goto('https://example.com');

  await page.waitForLoadState('networkidle');

  await expect(page.locator('.products')).toBeVisible();
```

4️⃣ page.waitForSelector()

📖 Description

Waits until an element appears on the page.

Useful for dynamic content loaded after page load.

🎯 When to Use

- AJAX content
- Lazy loading
- Dynamic elements

✅ Syntax

```
await page.waitForSelector(selector);
```

or

```
await page.waitForSelector(selector, {
      state: 'visible'
  });
```

Available States

State	Description
`'attached'`	Element exists in DOM
`'visible'`	Element is visible
`'hidden'`	Element becomes hidden
`'detached'`	Element removed from DOM

💻 Example

```
await page.click('#loadProducts');

  await page.waitForSelector('.product-card');

  await expect(page.locator('.product-card')).toHaveCount(10);
```

5 page.waitForTimeout()

📖 Description

Pauses execution for a fixed number of milliseconds.

⚠️ Avoid in Real Tests

Hard waits make tests:

- ❌ Slow
- ❌ Unstable
- ❌ Flaky

Prefer waiting for:

- locators
- assertions
- load states

🎯 Good Use Cases

- Debugging
- Demo videos
- Learning Playwright

✅ Syntax

```
await page.waitForTimeout(milliseconds);
```

💻 Example

```
await page.click('#submit');

  await page.waitForTimeout(3000);

  await page.screenshot({ path: 'result.png' });
```

6️⃣ page.waitForURL()

📖 Description

Waits until the page navigates to a specific URL.

Useful after login, form submission, or redirects.

🎯 When to Use

- Login success
- Checkout flow
- Redirect verification

✅ Syntax

```
await page.waitForURL(url);
```

💻 Example

```
await page.click('text=Login');

  await page.waitForURL('**/dashboard');

  await expect(page).toHaveURL(/dashboard/);
```

7️⃣ page.waitForFunction()

📖 Description

Waits until a JavaScript expression returns **true**.

Useful for waiting on custom conditions.

🎯 When to Use

- Custom JavaScript conditions
- Dynamic values
- Local storage updates
- Page variables

✅ Syntax

```
await page.waitForFunction(() => {
      return condition;
  });
```

💻 Example

```
await page.waitForFunction(() => {

      return document.title === 'Dashboard';

  });

  await expect(page).toHaveTitle('Dashboard');
```

8️⃣ page.waitForEvent()

📖 Description

Waits until a specific browser event occurs.

Commonly used for:

- downloads

- popups
- dialogs
- file chooser

🎯 When to Use

- File downloads
- New tabs
- Browser dialogs
- File upload windows

✅ Syntax

```
await page.waitForEvent(eventName);
```

Common Events

Event	Description
`'download'`	Wait for download
`'popup'`	Wait for new tab/window
`'dialog'`	Wait for alert/confirm/prompt
`'filechooser'`	Wait for file chooser

💻 Example (Popup)

```
const popupPromise = page.waitForEvent('popup');

  await page.click('text=Open Window');

  const popup = await popupPromise;

  await expect(popup).toHaveTitle(/New Page/);
```

💻 Example (Download)

```
const downloadPromise = page.waitForEvent('download');

  await page.click('text=Download');

  const download = await downloadPromise;

  console.log(await download.suggestedFilename());
```

🚀 Best Practices

✅ Prefer These

- ✔ `expect(locator).toBeVisible()`
- ✔ `locator.click()` (auto-waits)
- ✔ `locator.fill()` (auto-waits)
- ✔ `page.waitForURL()`
- ✔ `page.waitForSelector()`
- ✔ `page.waitForFunction()` (for custom conditions)

❌ Avoid These

- `page.waitForTimeout()` in production tests
- Excessive use of `networkidle`
- Waiting longer than necessary

💡 Which Wait Should I Use?

Scenario	Recommended Method
Wait until DOM is ready	`waitForLoadState('domcontentloaded')`
Wait until page fully loads	`waitForLoadState('load')`
Wait until API requests finish	`waitForLoadState('networkidle')` *(use cautiously)*
Wait for an element	`waitForSelector()`
Wait for URL change	`waitForURL()`
Wait for a custom JavaScript condition	`waitForFunction()`
Wait for popup/download/dialog	`waitForEvent()`
Pause execution temporarily	`waitForTimeout()` *(debugging only)*

🎯 Quick Summary

Method	Beginner Recommendation
⭐ `waitForLoadState('domcontentloaded')`	⭐⭐⭐⭐
⭐ `waitForLoadState('load')`	⭐⭐⭐⭐
⭐ `waitForLoadState('networkidle')`	⭐⭐
⭐ `waitForSelector()`	⭐⭐⭐⭐⭐
⭐ `waitForTimeout()`	⭐
⭐ `waitForURL()`	⭐⭐⭐⭐⭐
⭐ `waitForFunction()`	⭐⭐⭐⭐
⭐ `waitForEvent()`	⭐⭐⭐⭐

📌 Key Takeaways

- 🤖 Playwright automatically waits for most actions—explicit waits are only needed in specific scenarios.
- 🚀 Prefer **locators** and **assertions** over manual waits whenever possible.
- 🎯 Use `waitForSelector()` for dynamic elements and `waitForURL()` for navigation changes.
- 🧠 Use `waitForFunction()` for custom JavaScript conditions that cannot be handled by built-in waits.

- 📥 Use `waitForEvent()` for browser events like downloads, popups, dialogs, and file choosers.
- ⚠️ Avoid `waitForTimeout()` in production tests; reserve it for debugging or demonstrations.
- 🌐 Use `networkidle` sparingly, as waiting for specific UI changes or API responses usually produces more reliable tests.